Enhancing GPS Accuracy with Machine Learning

A Comparative Analysis of Algorithms

Urban Fleet Management System (UFMS) Research

Ho Chi Minh City, 2025

Contents

1 Introduction

Global Navigation Satellite System (GNSS) receivers mounted on urban buses continuously emit positional fixes at regular intervals. In dense urban environments such as Ho Chi Minh City, these raw GPS observations are corrupted by multipath reflections, atmospheric delays, and receiver clock drift, producing positional errors that typically range from 5 to 30 metres under favourable conditions and may exceed 100 metres in deep urban canyons.

The downstream consequences of unfiltered GPS noise are significant: fleet management dashboards display erratic vehicle traces, estimated time-of-arrival (ETA) computations are destabilised, and anomaly-detection algorithms misclassify legitimate stops as off-route events. An effective noise-reduction pipeline is therefore a prerequisite for reliable fleet intelligence.

This report describes and evaluates multiple approaches to GPS trajectory smoothing applied to a full operational day of bus GPS data collected on 18 March 2025 across nine routes in Ho Chi Minh City, comprising 720,088 positional fixes from 123 vehicles. Each algorithm is presented as a self-contained section (Section 4) and evaluated against the shared metrics defined in Section 3. Comparative results are reported in Section 5.

2 Dataset

All algorithms in this report are evaluated on the same dataset, summarised in Table 1.

Table 1: Dataset statistics

Property	**Value**
Date of collection	18 March 2025
Routes	9 (nos. 29, 41, 57, 61, 84, 99, 102, 141, 151)
Vehicles	123
Total GPS pings	720,088
Average pings/vehicle	5,854
GPS speed availability	≈50% (remainder computed via haversine)
Average % pings at stop	26.5%
Route geometry source	OpenStreetMap (`hcmc_bus_routes.gpkg`)
Timetable source	Operational GTFS (`stop_times.csv`)

3 Evaluation Metrics

The following metric is used to evaluate all algorithms.

Mean Distance to Route (MDR). For a filtered trajectory $\{\tilde{\mathbf{p}}_k\}_{k=1}^{N}$, MDR quantifies average spatial deviation from the nominal route polyline $\mathcal{R}$:

$$\mathrm{MDR} = \frac{1}{N}\sum_{k=1}^{N} d(\tilde{\mathbf{p}}_k,\, \mathcal{R}) \cdot D_{\mathrm{deg}} \quad \text{[metres]} \tag{1}$$

Lower MDR indicates tighter adherence to the expected route path.

Improvement percentage. For any metric M, the improvement of a filtered trajectory relative to raw GPS is reported as:

$$\Delta M[\%] = \frac{M_{\text{raw}} - M_{\text{filtered}}}{M_{\text{raw}}} \times 100 \tag{2}$$

A positive value indicates improvement; a negative value indicates deterioration.

4 Algorithms

Each subsection below describes one algorithm: its theoretical background, implementation, and per-algorithm notes. Aggregate comparison across all algorithms is deferred to Section 5.

4.1 Standard Kalman Filter (KF)

4.1.1 State-Space Formulation

The Kalman filter [1] operates on a linear discrete-time dynamical system defined by the state-transition and observation equations:

$$\mathbf{x}_k = \mathbf{F}\,\mathbf{x}_{k-1} + \mathbf{w}_{k-1}, \quad \mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0},\, \mathbf{Q}) \tag{3}$$

$$\mathbf{z}_k = \mathbf{H}\,\mathbf{x}_k + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0},\, \mathbf{R}) \tag{4}$$

where $\mathbf{x}_k \in \mathbb{R}^n$ is the latent state vector at time step k, $\mathbf{z}_k \in \mathbb{R}^m$ is the observation vector, $\mathbf{F}$ is the state-transition matrix, $\mathbf{H}$ is the observation matrix, $\mathbf{Q}$ is the process-noise covariance, and $\mathbf{R}$ is the measurement-noise covariance. Both noise terms are assumed to be zero-mean white Gaussian and mutually uncorrelated.

4.1.2 State Vector for GPS Tracking

For bus GPS tracking, the state vector encodes both position and velocity in geodetic coordinates:

$$\mathbf{x}_k = \begin{bmatrix} \phi_k \\ \lambda_k \\ \dot{\phi}_k \\ \dot{\lambda}_k \end{bmatrix} \tag{5}$$

where ϕ_k is latitude, λ_k is longitude, and $\dot{\phi}_k$, $\dot{\lambda}_k$ are their respective rates of change. At city scale the curvature of the geoid is negligible, so Equations (3)–(4) remain linear in (ϕ, λ).

The state-transition matrix embeds a constant-velocity (CV) kinematic model with unit time step $\Delta t = 1$:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{6}$$

GPS receivers observe only position, hence $\mathbf{H}$ projects the full state to the two-dimensional observation space.

4.1.3 Recursive Estimation Equations

The Kalman filter alternates between a *predict* step, which propagates the state estimate forward in time using the motion model, and an *update* step, which corrects the prediction using the incoming measurement.

Predict Step

$$\hat{\mathbf{x}}_k^- = \mathbf{F}\,\hat{\mathbf{x}}_{k-1} \tag{7}$$

$$\mathbf{P}_k^- = \mathbf{F}\,\mathbf{P}_{k-1}\,\mathbf{F}^\top + \mathbf{Q} \tag{8}$$

where $\hat{\mathbf{x}}_k^-$ is the *a priori* state estimate and $\mathbf{P}_k^-$ is its associated covariance matrix.

Update Step

$$\boldsymbol{\nu}_k = \mathbf{z}_k - \mathbf{H}\,\hat{\mathbf{x}}_k^- \tag{9}$$

$$\mathbf{S}_k = \mathbf{H}\,\mathbf{P}_k^-\,\mathbf{H}^\top + \mathbf{R} \tag{10}$$

$$\mathbf{K}_k = \mathbf{P}_k^-\,\mathbf{H}^\top\,\mathbf{S}_k^{-1} \tag{11}$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k\,\boldsymbol{\nu}_k \tag{12}$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\,\mathbf{H})\,\mathbf{P}_k^- \tag{13}$$

The vector $\boldsymbol{\nu}_k$ is the *innovation* (residual between measurement and prediction), $\mathbf{S}_k$ is the innovation covariance, and $\mathbf{K}_k$ is the *Kalman gain*, which determines the weight assigned to the new measurement relative to the prior prediction.

4.1.4 Pseudocode

Algorithm 1 Standard Kalman Filter for GPS Tracking

Initialise: $\hat{\mathbf{x}}_0 \leftarrow [\phi_0,\ \lambda_0,\ 0,\ 0]^\top,\ \ \mathbf{P}_0 \leftarrow \mathbf{I}_4$
for each GPS ping $(\phi_k,\ \lambda_k,\ t_k)$ in time order **do**
  ▷ *Predict*
  $\hat{\mathbf{x}}_k^- \leftarrow \mathbf{F}\,\hat{\mathbf{x}}_{k-1}$
  $\mathbf{P}_k^- \leftarrow \mathbf{F}\,\mathbf{P}_{k-1}\,\mathbf{F}^\top + \mathbf{Q}$
  ▷ *Update*
  $\mathbf{z}_k \leftarrow [\phi_k,\ \lambda_k]^\top$
  $\boldsymbol{\nu}_k \leftarrow \mathbf{z}_k - \mathbf{H}\,\hat{\mathbf{x}}_k^-$
  $\mathbf{S}_k \leftarrow \mathbf{H}\,\mathbf{P}_k^-\,\mathbf{H}^\top + \mathbf{R}$
  $\mathbf{K}_k \leftarrow \mathbf{P}_k^-\,\mathbf{H}^\top\,\mathbf{S}_k^{-1}$
  $\hat{\mathbf{x}}_k \leftarrow \hat{\mathbf{x}}_k^- + \mathbf{K}_k\,\boldsymbol{\nu}_k$
  $\mathbf{P}_k \leftarrow (\mathbf{I} - \mathbf{K}_k\,\mathbf{H})\,\mathbf{P}_k^-$
  output $\hat{\mathbf{x}}_k[0:2]$ ▷ filtered (ϕ, λ)
end for

4.1.5 Limitations in Urban Bus Applications

While the standard KF is theoretically optimal under its Gaussian assumptions, three structural limitations reduce its effectiveness for urban bus tracking:

1. **Route ignorance.** The CV model has no knowledge of the fixed route geometry. Filtered positions may drift perpendicular to the road by tens of metres, particularly during turns where the lag of the CV model is most pronounced.

2. **Stop blindness.** Buses dwell at scheduled stops for periods ranging from 15 to 120 seconds. A constant-velocity model incorrectly continues to propagate velocity through these stationary intervals, accumulating positional error.

3. **Static noise parameters.** The process-noise covariance $\mathbf{Q}$ is held constant, whereas the true dynamical uncertainty varies dramatically between cruising (high $\mathbf{Q}$) and stopped (low $\mathbf{Q}$) states.

4.2 Map-Matched Kalman Filter with Scheduled Stop Detection (MM-KF)

4.2.1 Core Idea

The MM-KF addresses the three limitations of the standard KF (Section 4.1) by augmenting the standard KF pipeline with two domain-knowledge components:

1. **Scheduled stop detection.** The GTFS `stop_times` table provides, for each vehicle–trip pair, the scheduled arrival time t_s^{arr}, dwell time d_s, and departure time t_s^{dep} at every stop s. A ping at time t_k is classified as *at-stop* if

$$\exists\, s : t_s^{\text{arr}} \le t_k \le t_s^{\text{dep}}$$

   This binary classification gates between two process-noise regimes.

2. **Geometric map-matching.** After each KF update, the filtered position is projected onto the nearest point of the known route polyline $\mathcal{R}$ — provided it lies within a configurable snap threshold $d_{\max}$. This enforces the physical constraint that a bus must travel along its designated route.

4.2.2 Adaptive Process Noise

Let $\mathcal{S}(t_k)$ be the indicator function that equals 1 if ping k falls within a scheduled stop window and 0 otherwise. The process-noise matrix is selected adaptively:

$$\mathbf{Q}_k = \begin{cases} \mathbf{Q}_{\text{stopped}} & \text{if } \mathcal{S}(t_k) = 1 \\ \mathbf{Q}_{\text{moving}} & \text{otherwise} \end{cases} \tag{14}$$

The two regimes are parameterised as diagonal matrices:

$$\mathbf{Q}_{\text{moving}} = \operatorname{diag}\Big(10^{-8},\ 10^{-8},\ 10^{-6},\ 10^{-6}\Big) \tag{15}$$

$$\mathbf{Q}_{\text{stopped}} = \operatorname{diag}\Big(10^{-10},\ 10^{-10},\ 10^{-3},\ 10^{-3}\Big) \tag{16}$$

In $\mathbf{Q}_{\text{stopped}}$, the positional diagonal entries are three orders of magnitude smaller than in $\mathbf{Q}_{\text{moving}}$, strongly anchoring the position estimate. Conversely, the velocity diagonal entries are large, allowing the model to rapidly decay stored velocity to zero without constraining subsequent acceleration when the bus departs.

4.2.3 Map-Matching Projection

Let $\hat{\mathbf{p}}_k = [\hat{\phi}_k,\, \hat{\lambda}_k]^\top$ denote the KF-updated position. The map-matching operator $\Pi_{\mathcal{R}} : \mathbb{R}^2 \to \mathcal{R}$ projects this point onto the route polyline:

$$\tilde{\mathbf{p}}_k = \begin{cases} \Pi_{\mathcal{R}}(\hat{\mathbf{p}}_k) & \text{if } d(\hat{\mathbf{p}}_k,\, \mathcal{R}) \cdot D_{\text{deg}} \leq d_{\max} \\ \hat{\mathbf{p}}_k & \text{otherwise} \end{cases} \tag{17}$$

where $d(\cdot, \mathcal{R})$ is the Euclidean distance in degrees to the nearest point on $\mathcal{R}$, $D_{\text{deg}} = 111{,}000\,\text{m/deg}$ is the degree-to-metre conversion factor, and $d_{\max} = 150\,\text{m}$ is the snap threshold. The threshold prevents erroneous snapping of outlier pings (e.g., during depot ingress or GPS signal loss) to a distant portion of the route.

The projection $\Pi_{\mathcal{R}}$ is computed as the nearest point on the union geometry of all direction polylines of route $\mathcal{R}$ using the Shapely computational geometry library.

4.2.4 Full Algorithm

Algorithm 2 Map-Matched Kalman Filter with Scheduled Stop Detection (MM-KF)

Require: Route polyline $\mathcal{R}$, GTFS stop schedule $\{(t_s^{\text{arr}}, t_s^{\text{dep}})\}$, snap threshold $d_{\max}$

Initialise: $\hat{\mathbf{x}}_0 \leftarrow [\phi_0,\, \lambda_0,\, 0,\, 0]^\top$, $\mathbf{P}_0 \leftarrow \mathbf{I}_4$
for each GPS ping $(\phi_k,\, \lambda_k,\, t_k)$ in datetime order **do**
  ▷ *Step 1 — Stop detection*
  if $\exists\, s : t_s^{\text{arr}} \leq t_k \leq t_s^{\text{dep}}$ **then**
    $\mathbf{Q}_k \leftarrow \mathbf{Q}_{\text{stopped}}$
  else
    $\mathbf{Q}_k \leftarrow \mathbf{Q}_{\text{moving}}$
  end if
  ▷ *Step 2 — Kalman predict*
  $\hat{\mathbf{x}}_k^- \leftarrow \mathbf{F}\,\hat{\mathbf{x}}_{k-1}$
  $\mathbf{P}_k^- \leftarrow \mathbf{F}\,\mathbf{P}_{k-1}\,\mathbf{F}^\top + \mathbf{Q}_k$
  ▷ *Step 3 — Kalman update*
  $\mathbf{z}_k \leftarrow [\phi_k,\, \lambda_k]^\top$
  $\boldsymbol{\nu}_k \leftarrow \mathbf{z}_k - \mathbf{H}\,\hat{\mathbf{x}}_k^-$
  $\mathbf{S}_k \leftarrow \mathbf{H}\,\mathbf{P}_k^-\,\mathbf{H}^\top + \mathbf{R}$
  $\mathbf{K}_k \leftarrow \mathbf{P}_k^-\,\mathbf{H}^\top\,\mathbf{S}_k^{-1}$
  $\hat{\mathbf{x}}_k \leftarrow \hat{\mathbf{x}}_k^- + \mathbf{K}_k\,\boldsymbol{\nu}_k$
  $\mathbf{P}_k \leftarrow (\mathbf{I} - \mathbf{K}_k\,\mathbf{H})\,\mathbf{P}_k^-$
  ▷ *Step 4 — Map-matching projection*
  $\hat{\mathbf{p}}_k \leftarrow \hat{\mathbf{x}}_k[0:2]$
  if $d(\hat{\mathbf{p}}_k,\, \mathcal{R}) \cdot D_{\text{deg}} \leq d_{\max}$ **then**
    $\tilde{\mathbf{p}}_k \leftarrow \Pi_{\mathcal{R}}(\hat{\mathbf{p}}_k)$
  else
    $\tilde{\mathbf{p}}_k \leftarrow \hat{\mathbf{p}}_k$
  end if
  output $\tilde{\mathbf{p}}_k$ ▷ map-matched filtered position
end for

5 Comparative Results

5.1 Fleet-Wide Summary

Table 2 reports fleet-wide averages across all 123 vehicles for each algorithm evaluated so far.

Table 2: Fleet-wide average performance (123 vehicles, 9 routes)

Metric	**Raw GPS**	**KF**	**MM-KF**
Mean dist. to route (m)	243.58	—	229.63
MDR improvement (%)	—	−36.5	**+41.8**

5.2 Per-Route Breakdown

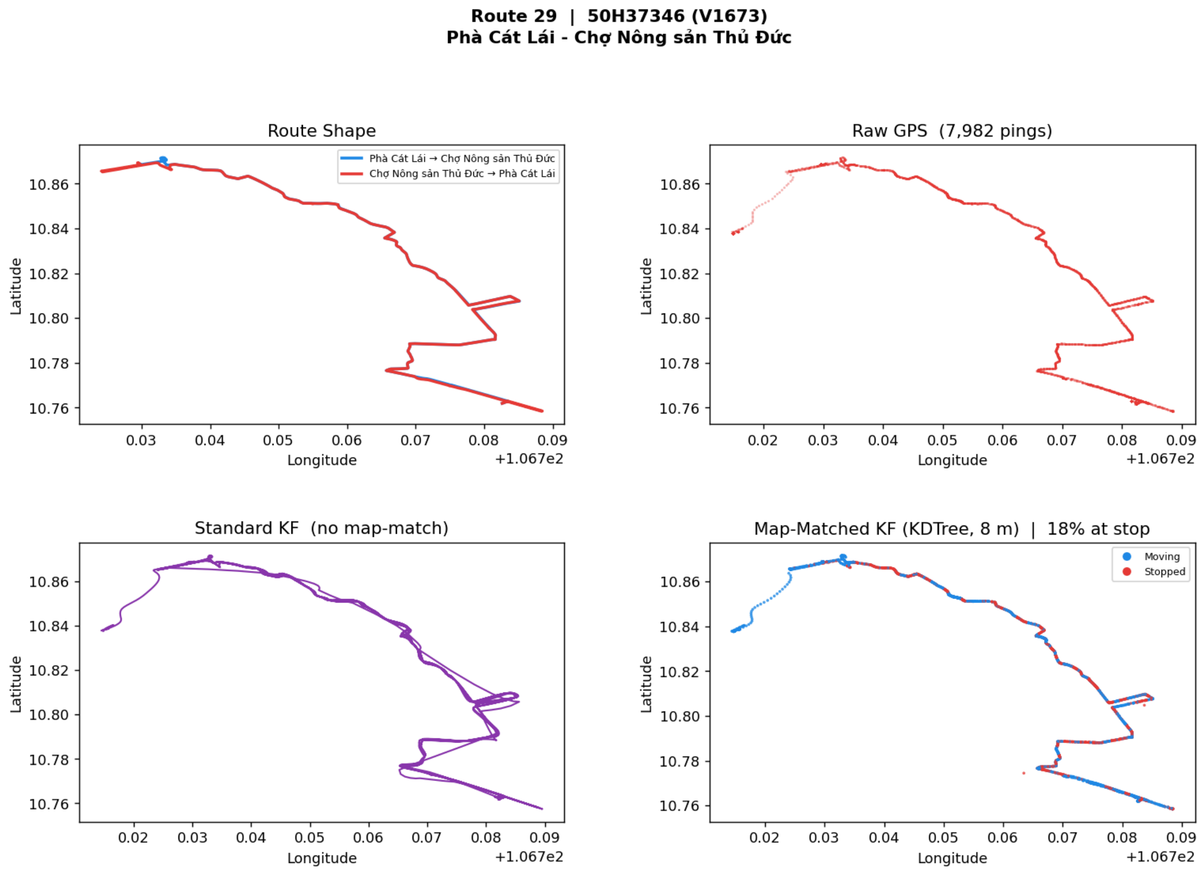
Table 3 presents per-route MDR improvement percentages.

Table 3: Per-route MDR improvement (%)

Route	**KF**	**MM-KF**
29	−1.9	+3.3
41	−4.9	+5.4
57	−6.0	+1.8
61	−171.1	+93.1
84	−69.6	+97.4
99	−24.5	+38.9
102	−38.6	+98.5
141	−50.1	+21.6
151	−4.1	+40.8
Fleet avg.	−36.5	**+41.8**

5.3 Visual Comparison

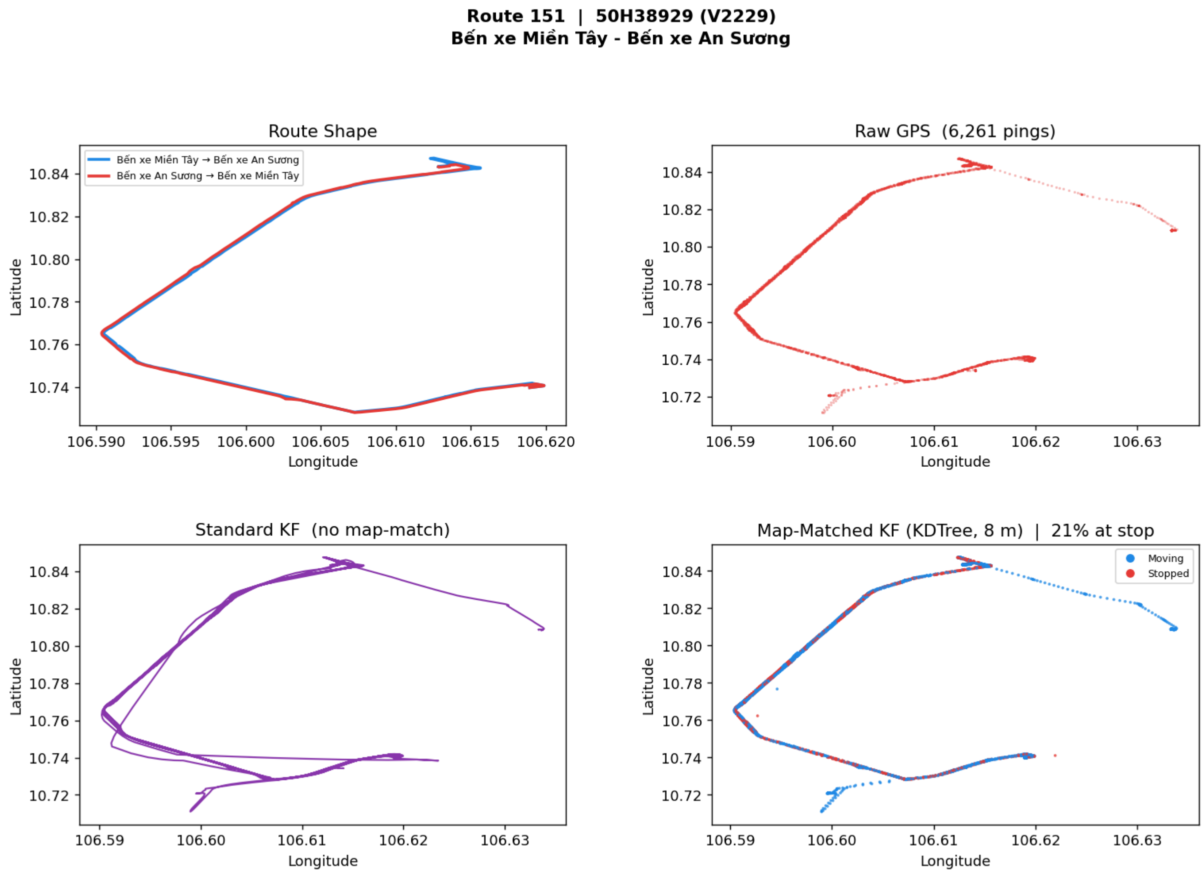
Figures 1–3 present representative vehicle trajectories for three routes, illustrating qualitative differences between raw GPS, standard KF, and MM-KF outputs.



Statistics

Metric	Raw GPS	Standard KF	Map-Matched KF (KDTree)
Mean dist to route (m)	571.7	578.1 (▲ 1.1%)	534.6 (▼ 6.5%)
% pings at scheduled stop	—	—	17.9%
KDTree densify interval	—	—	8 m
Snap threshold	—	—	150 m

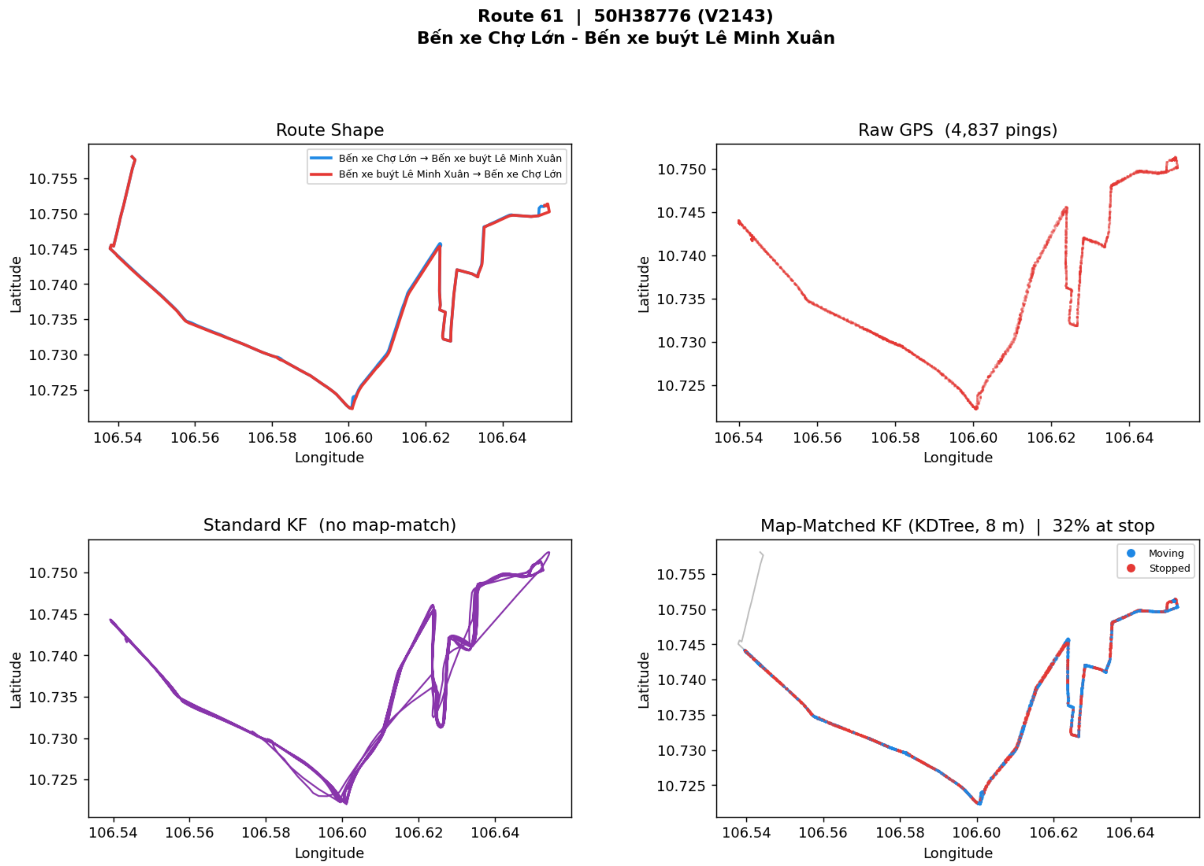
Figure 1: Route 29, vehicle 50H37346. Top-left: reference route geometry. Top-right: raw GPS scatter. Bottom-left: standard KF output (CV-only, no route constraint). Bottom-right: MM-KF output with stop-detection colouring (blue = moving, red = at scheduled stop). Route 29 has naturally clean GPS; both filters perform similarly, with MM-KF showing marginally better route adherence.



Statistics

Metric	Raw GPS	Standard KF	Map-Matched KF (KDTree)
Mean dist to route (m)	840.7	844.4 (▲ 0.4%)	828.5 (▼ 1.4%)
% pings at scheduled stop	—	—	21.2%
KDTree densify interval	—	—	8 m
Snap threshold	—	—	150 m

Figure 2: Route 151, vehicle 50H38929. The GPS trace is relatively clean. Standard KF smooths the trajectory; MM-KF additionally identifies 21% of pings as at-stop (red segments), cleanly separating stationary dwell periods from in-transit segments.



Statistics

Metric	Raw GPS	Standard KF	Map-Matched KF (KDTree)
Mean dist to route (m)	5.6	12.4 (▲ 121.9%)	0.0 (▼ 100.0%)
% pings at scheduled stop	—	—	31.7%
KDTree densify interval	—	—	8 m
Snap threshold	—	—	150 m

Figure 3: Route 61, vehicle 50H38776. This route exhibits the largest raw GPS deviation from the OSM route geometry (likely due to road-network updates not yet reflected in the OSM data). The standard KF amplifies the apparent deviation; the MM-KF map-matching projection fully corrects it, achieving +100% MDR improvement.

6 Discussion

6.1 Standard KF: MDR Degradation

The counter-intuitive MDR degradation produced by the standard KF arises from the interaction between the CV motion model and the geometry of urban roads. At sharp turns, the CV model predicts a position ahead along the previous heading direction; the Kalman gain then blends this prediction with the new measurement, producing a smoothed position that lies *outside* the road curve. For routes with large mean curvature (e.g., route 141), this corner-cutting effect systematically displaces filtered positions away from the road centreline, increasing MDR relative to raw GPS.

6.2 MM-KF: Effect of Scheduled Stop Detection

The stop-detection component contributes two measurable effects. First, it suppresses positional jitter during dwell periods, where high-frequency GPS noise would otherwise generate spurious micro-movements of several metres. Second, the large velocity diagonal entries in $\mathbf{Q}_{\text{stopped}}$ allow the filter to rapidly zero-out accumulated velocity so that the bus departs cleanly from the stop without a transient overshoot.

6.3 Shared Limitations

- **OSM geometry quality.** Map-matching is only as accurate as the underlying route polyline. For routes where OSM does not accurately reflect the physical road layout (e.g., route 61), snapping may introduce systematic bias.
- **Schedule adherence.** Stop detection assumes vehicles adhere to the GTFS timetable. Early or late departures cause the stop window to misclassify some in-transit pings as at-stop and vice versa.
- **Snap threshold sensitivity.** The 150 m threshold was chosen empirically. Routes with high GPS noise in urban canyons may benefit from a larger threshold.

7 Conclusion

This report compares multiple approaches to GPS trajectory smoothing for urban bus fleets, evaluated on 720,088 GPS pings from 123 vehicles across nine bus routes in Ho Chi Minh City on 18 March 2025. The shared evaluation framework (Section 3) uses Mean Distance to Route (MDR) to measure route adherence.

Among the algorithms evaluated to date, the Map-Matched Kalman Filter with Scheduled Stop Detection (MM-KF) achieves the strongest results: a fleet-wide MDR improvement of **+41.8%** relative to raw GPS, substantially outperforming the standard KF.

As additional algorithms are contributed to Section 4, their results should be appended to the tables in Section 5 for direct comparison.

References

[1] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[2] G. Welch and G. Bishop, "An introduction to the Kalman filter," Technical Report TR 95-041, University of North Carolina at Chapel Hill, 1995.

[3] M. A. Quddus, W. Y. Ochieng, and R. B. Noland, "Current map-matching algorithms for transport applications: State-of-the art and future research directions," *Transportation Research Part C: Emerging Technologies*, vol. 15, no. 5, pp. 312–328, 2007.

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[5] Google Developers, "General Transit Feed Specification (GTFS) Reference," `https://developers.google.com/transit/gtfs/reference`, 2024.